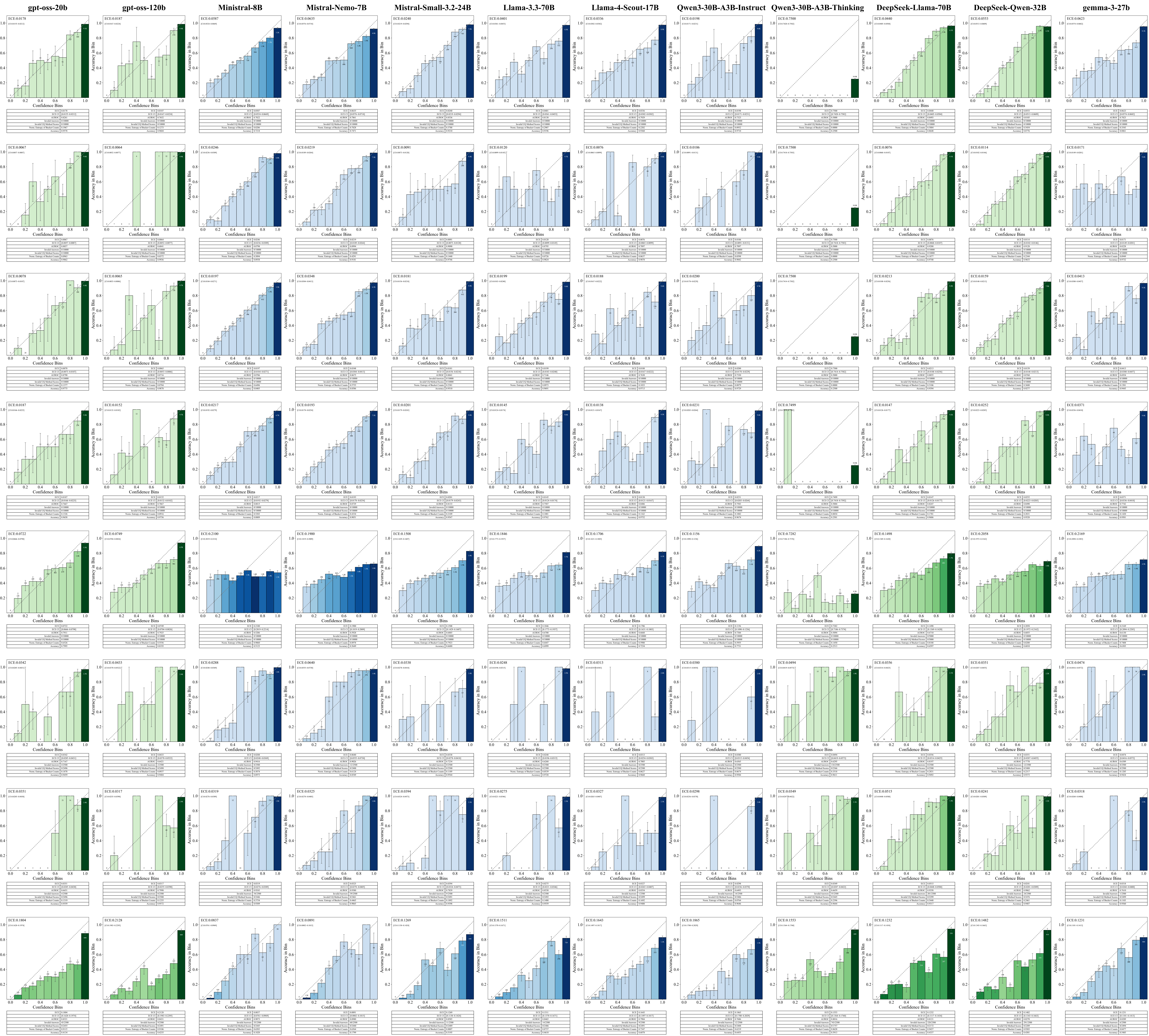
Calibration Plots for Uncertainty Metric Frequency of Answer



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0178	0.0187	0.0587	0.0635	0.0240	0.0401	0.0336	0.0198	0.7500	0.0440	0.0553	0.0623
ECE CI	[0.0155–0.0212]	[0.0167–0.0224]	[0.0524–0.0669]	[0.0576–0.0718]	[0.0219–0.0294]	[0.0361–0.0453]	[0.0302–0.0382]	[0.0171–0.0231]	[0.7420–0.7582]	[0.0405–0.0504]	[0.0513–0.0609]	[0.0574–0.0682]
AUROC	0.8261	0.7652	0.7822	0.7861	0.8536	0.8138	0.7929	0.7323	0.5000	0.8493	0.8105	0.7823
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1967	0.1273	0.8286	0.7828	0.3788	0.2907	0.2282	0.0932	0.0000	0.5884	0.5454	0.3543
Accuracy	0.9578	0.9684	0.7218	0.7471	0.9210	0.9196	0.9360	0.9710	0.2500	0.8648	0.8776	0.8861

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0067	0.0064	0.0246	0.0219	0.0091	0.0120	0.0076	0.0106	0.7500	0.0076	0.0114	0.0171
ECE CI	[0.0057–0.0087]	[0.0052–0.0077]	[0.0236–0.0309]	[0.0189–0.0264]	[0.0073–0.0120]	[0.0099–0.0145]	[0.0065–0.0099]	[0.0091–0.0131]	[0.7418–0.7583]	[0.0068–0.0107]	[0.0102–0.0146]	[0.0149–0.0201]
AUROC	0.8857	0.6841	0.8798	0.8994	0.9090	0.8795	0.7897	0.7097	0.5000	0.9288	0.9128	0.8228
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0963	0.0372	0.5094	0.4291	0.1440	0.0726	0.0637	0.0598	0.0000	0.1877	0.2344	0.0949
Accuracy	0.9862	0.9936	0.8938	0.9181	0.9746	0.9824	0.9878	0.9842	0.2500	0.9748	0.9653	0.9732

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0078	0.0065	0.0197	0.0348	0.0181	0.0199	0.0188	0.0200	0.7500	0.0213	0.0159	0.0413
ECE CI	[0.0073–0.0107]	[0.0053–0.0086]	[0.0184–0.0271]	[0.0304–0.0413]	[0.0156–0.0218]	[0.0183–0.0240]	[0.0167–0.0222]	[0.0178–0.0229]	[0.7414–0.7582]	[0.0188–0.0256]	[0.0148–0.0213]	[0.0380–0.0457]
AUROC	0.8790	0.8716	0.8706	0.8675	0.8841	0.7146	0.7630	0.7330	0.5000	0.8855	0.8768	0.6934
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1377	0.0784	0.6496	0.5755	0.2381	0.0973	0.1033	0.0879	0.0000	0.3106	0.3680	0.1501
Accuracy	0.9775	0.9878	0.8405	0.8586	0.9510	0.9691	0.9713	0.9729	0.2500	0.9394	0.9277	0.9445

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0187	0.0152	0.0217	0.0193	0.0201	0.0145	0.0138	0.0231	0.7499	0.0147	0.0252	0.0371
ECE CI	[0.0166–0.0225]	[0.0132–0.0182]	[0.0192–0.0279]	[0.0178–0.0254]	[0.0179–0.0243]	[0.0124–0.0174]	[0.0121–0.0167]	[0.0203–0.0264]	[0.7418–0.7592]	[0.0126–0.0177]	[0.0223–0.0285]	[0.0336–0.0410]
AUROC	0.7667	0.7867	0.8815	0.8729	0.8133	0.7587	0.8486	0.7583	0.5004	0.8247	0.8490	0.7742
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1326	0.0916	0.5283	0.4334	0.2109	0.1001	0.1162	0.1082	0.0034	0.1701	0.2831	0.1420
Accuracy	0.9658	0.9756	0.8809	0.9051	0.9547	0.9762	0.9752	0.9674	0.2501	0.9686	0.9520	0.9501

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0722	0.0749	0.2100	0.1900	0.1508	0.1846	0.1706	0.1156	0.7282	0.1498	0.2058	0.2169
ECE CI	[0.0666–0.0798]	[0.0704–0.0826]	[0.2019–0.2214]	[0.1819–0.2009]	[0.1429–0.1607]	[0.1773–0.1937]	[0.1631–0.1803]	[0.1090–0.1236]	[0.7186–0.7376]	[0.1369–0.1620]	[0.1972–0.2162]	[0.2094–0.2283]
AUROC	0.7911	0.7925	0.5206	0.5928	0.6885	0.6700	0.6660	0.7384	0.5094	0.6710	0.6055	0.6130
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.6526	0.5556	0.9060	0.8978	0.7923	0.6447	0.5980	0.5955	0.1450	0.8188	0.8208	0.7408
Accuracy	0.7995	0.8193	0.5123	0.5649	0.6680	0.6999	0.7234	0.7731	0.2513	0.6597	0.6010	0.6303

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0342	0.0433	0.0288	0.0640	0.0338	0.0248	0.0313	0.0380	0.0494	0.0356	0.0351	0.0474
ECE CI	[0.0285–0.0431]	[0.0370–0.0522]	[0.0248–0.0369]	[0.0555–0.0730]	[0.0278–0.0418]	[0.0196–0.0315]	[0.0254–0.0383]	[0.0315–0.0454]	[0.0419–0.0573]	[0.0314–0.0425]	[0.0287–0.0455]	[0.0416–0.0572]
AUROC	0.7167	0.6421	0.9416	0.9026	0.7234	0.6268	0.7005	0.6565	0.6295	0.8197	0.7770	0.6389
Invalid Answers	4/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	134/2500	0/2500	12/2500	0/2500
Invalid UQ Method Scores	0/2496	0/2498	0/2492	0/2488	0/2500	0/2500	0/2500	0/2500	0/2366	0/2500	0/2488	0/2500
Norm. Entropy of Bucket Counts	0.1878	0.0937	0.5074	0.6124	0.1189	0.0539	0.0627	0.0674	0.3518	0.2831	0.2317	0.1285
Accuracy	0.9459	0.9484	0.8973	0.8549	0.9544	0.9728	0.9660	0.9584	0.9611	0.9492	0.9373	0.9424

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0351	0.0317	0.0319	0.0325	0.0394	0.0275	0.0327	0.0298	0.0349	0.0515	0.0241	0.0318
ECE CI	[0.0285–0.0430]	[0.0255–0.0390]	[0.0276–0.0389]	[0.0278–0.0405]	[0.0324–0.0473]	[0.0221–0.0346]	[0.0265–0.0407]	[0.0236–0.0370]	[0.0287–0.0422]	[0.0448–0.0588]	[0.0201–0.0309]	[0.0268–0.0400]
AUROC	0.6651	0.7998	0.9243	0.9380	0.7839	0.8530	0.8554	0.6493	0.6659	0.9358	0.9062	0.7610
Invalid Answers	4/2500	0/2500	54/2500	39/2500	9/2500	8/2500	1/2500	14/2500	55/2500	201/2500	24/2500	1/2500
Invalid UQ Method Scores	0/2496	0/2500	0/2446	0/2461	0/2491	0/2492	0/2499	0/2486	0/2445	0/2299	0/2476	0/2499
Norm. Entropy of Bucket Counts	0.1319	0.1255	0.3754	0.4465	0.1892	0.1400	0.1692	0.0764	0.2396	0.5448	0.2461	0.1103
Accuracy	0.9559	0.9572	0.9309	0.9065	0.9398	0.9559	0.9400	0.9646	0.9644	0.9317	0.9487	0.9560

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.1804	0.2128	0.0837	0.0891	0.1269	0.1511	0.1643	0.1865	0.1553	0.1232	0.1482	0.1231
ECE CI	[0.1620–0.1974]	[0.1983–0.2295]	[0.0761–0.0969]	[0.0802–0.1015]	[0.1130–0.1424]	[0.1370–0.1671]	[0.1497–0.1817]	[0.1700–0.2029]	[0.1364–0.1760]	[0.1117–0.1434]	[0.1343–0.1663]	[0.1101–0.1413]
AUROC	0.8321	0.8431	0.9073	0.8980	0.8585	0.8465	0.7984	0.7686	0.8016	0.8121	0.8270	0.8289
Invalid Answers	45/2500	9/2500	55/2500	60/2500	11/2500	14/2500	9/2500	12/2500	743/2500	328/2500	291/2500	6/2500
Invalid UQ Method Scores	0/2455	0/2491	0/2445	0/2440	0/2489	0/2486	0/2491	0/2488	0/1757	0/2172	0/2209	0/2494
Norm. Entropy of Bucket Counts	0.9115	0.9106	0.6583	0.6564	0.8807	0.8737	0.7347	0.6219	0.8628	0.9437	0.9303	0.8477
Accuracy	0.4134	0.4255	0.1820	0.1799	0.5195	0.4610	0.6078	0.6455	0.5913	0.4337	0.4301	0.5485